



First Semester 2026-27
Course Handout Part II

Date: 31st July 2026

In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number: ME F415
Course Title: Gas Dynamics
Instructor-in-Charge: Biswadip Shome (biswadip.shome@pilani.bits-pilani.ac.in)
Team of Instructors: N/A

1. Course description:

This is a first-level course offered to advanced first degree and higher degree students to understand the essential concepts in compressible flows. The course contents include Introduction to Gas Dynamics, Basic equations for compressible flow, Flow with friction and heat addition, Normal shock, Quasi one-dimensional flow, Oblique shock, Prandtl-Meyer Flow, Potential equation for compressible flow, Linearized flow solution, and Hypersonic flows.

2. Scope and Objective:

- a. Introduce students to important concepts and mathematical formulations of Gas Dynamics
- b. Enable students to formulate and solve engineering problems in Gas Dynamics

3. Course Learning Outcomes:

Upon successful completion of this course, the students will have the following learning outcomes:

- a. Develop a broad understanding of the basic concepts and theories in gas dynamics
- b. Develop an in-depth understanding of various gas dynamic concepts such as flows with heat addition and friction, normal and oblique shock waves, Prandtl-Meyer expansion waves, Quasi 1D flow in nozzles and diffusers, Linearized solution theory and applications, and Hypersonic flows

4. Textbooks and Reference Books:

- TB:** Ethirajan Rathakrishnan, Gas Dynamics, 7th Edition, Prentice, PHI Learning, 2020.
- RB:** John D. Anderson Jr., Modern Compressible Flow: With Historical Perspective, 4th Edition, McGraw Hill Education, 2021.

5. Method of Conduct of the Course:

The course will be conducted using a combination of instructor-centred and learner-centred methods. The former would involve instructor-led class lectures and problem solving exercises while the latter



would involve student-led collaborative group projects using simulation for experiential learning, technical report writing, and presentations.

6. Course Plan:

Week	Module	Lecture	Reference	Learning Outcome
1	M1: Introduction to Gas Dynamics	1-2	Ch. 1 TB RB Class slides	Understanding of introductory concepts and applications of gas dynamics principles
2-3	M2: Basic Concepts for Compressible Flow	3-8	Ch. 2-4 TB RB Class slides	Understand mathematical and thermodynamic formulations for compressible flows
3-4	M3: Flow with Heat Addition (Rayleigh Flow)	9-10	Ch. 11 TB RB Class slides	- Understand and analyse adiabatic compressible gas flow with friction in constant area ducts - Apply Rayleigh flow principles to practical engineering problems
4	M4: Flow with Friction (Fanno Flow)	11-12	Ch. 11 TB RB Class slides	- Understand and analyse compressible gas flow with heat addition in constant area ducts - Apply Fanno flow principles to practical engineering problems
5-6	M5: Normal Shock Waves	13-18	Ch. 5 TB RB Class slides	- Comprehension of Normal shock formation, their properties, and their effects on flow characteristics - Application of Normal shock theory to practical engineering scenarios in supersonic inlets, nozzles and diffusers
7-8	M6: Quasi One-Dimensional Flow	19-24	Ch. 4 TB RB Class slides	- Understanding of inviscid compressible adiabatic gas flow through variable area passages - Application of Quasi-1D flow theory to nozzles and diffusers
9	M7: Oblique Shock and Expansion Waves	25-27	Ch. 6 TB RB Class slides	Ability to identify, analyse, and apply Oblique shock wave theory to supersonic flows
10	M8: Prandtl-Meyer Flows	28-30	Ch. 10 TB RB Class Slides	Understand expansion of supersonic flow around convex corners



Week	Module	Lecture	Reference	Learning Outcome
				Ability to calculate property changes using the Prandtl-Meyer function
11-12	M9: Potential Equations for Compressible Flow	31-34	Ch. 7 TB RB Class Slides	Understand governing equations, recognize their limitations, and apply them to solve problems related to steady, irrotational, and isentropic compressible flows
12-13	M10: Linearized Solutions	35-38	Ch. 8 TB RB Class Slides	Understand and apply linearized solutions to problems like subsonic and supersonic flow over aerofoils
14	M11: Hypersonic Flows	39-40	RB Class Slides	Ability to describe Hypersonic flow physics, differentiate between hypersonic and supersonic flows and apply hypersonic flow theory to engineering applications

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Comments
Quiz	20 minutes	15%	TBA	- Best 04 out of 05 - Open text book¹ - Open class notes
Course project	-	15%	Report due date 15/11/2026	- Group project - Open resources
Mid-Semester exam	90 minutes	30%	08/10/26 (FN1)	- Open text book¹ - Open class notes
Comprehensive exam	180 minutes	40%	11/12/26 (FN)	- Open text book¹ - Open class notes

Notes:

- Only prescribed text book (Gas Dynamics by Ethirajan Rathakrishnan) and hand-written class notes are allowed.
- Printed material, other books, photocopied materials, loose sheets are strictly prohibited. Use of any unapproved materials during the examination will be considered as malpractice and will invite strict disciplinary action as per the institute rules.



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3. Course project shall be taken in a group of maximum 3-4 members. The list of project topics will be shared by the IC. The evaluation of course project will be based on the report submitted and a presentation may be required. Project report submitted after due date will not be graded.

8. Chamber Consultation Hours: FD-II Room 2228, Wednesdays 3:00 PM to 4:00 PM.

9. Make-up Policy: Make up for Mid-semester and Comprehensive Examination will be conducted based on the requests received and as per the guidelines by AUGS/AGSR Division. No make-ups will be conducted for Quizzes.

10. Attendance Policy: As per guidelines by AUGS/AGSR Division.

11. Notices: Notices pertaining to this course will be displayed on Nalanda course page or be sent through email.

Prof. Biswadip Shome
Instructor-in-Charge ME F415